# Multiple Regression

ITAO 20200: Statistical Inference in Business

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```r
# Reading in the dataset
MedCosts <- read.csv("https://www.crc.nd.edu/~jwaddell/ITAO20200/Data/MedicalCosts.csv")
```

### Basic Descriptive Statistics

```r
library(psych)
describe(MedCosts[,c("age", "bmi", "children", "charges")], fast = TRUE)
```

```
##          vars    n     mean       sd     min      max    range     se
## age         1 1338    39.21    14.05   18.00    64.00    46.00   0.38
## bmi         2 1338    30.66     6.10   15.96    53.13    37.17   0.17
## children    3 1338     1.09     1.21    0.00     5.00     5.00   0.03
## charges     4 1338 13270.42 12110.01 1121.87 63770.43 62648.55 331.07
```

### Correlations

```r
library(psych)
corr.test(MedCosts[, c("age", "sex", "bmi", "children",
                       "smoker", "charges")], adjust = "none")
```

```
## Call:corr.test(x = MedCosts[, c("age", "sex", "bmi", "children", "smoker",
##     "charges")], adjust = "none")
## Correlation matrix
##            age   sex  bmi children smoker charges
## age       1.00 -0.02 0.11     0.04  -0.03    0.30
## sex      -0.02  1.00 0.05     0.02   0.08    0.06
## bmi       0.11  0.05 1.00     0.01   0.00    0.20
## children  0.04  0.02 0.01     1.00   0.01    0.07
## smoker   -0.03  0.08 0.00     0.01   1.00    0.79
## charges   0.30  0.06 0.20     0.07   0.79    1.00
## Sample Size
## [1] 1338
## Probability values (Entries above the diagonal are adjusted for multiple tests.)
##           age  sex  bmi children smoker charges
## age      0.00 0.45 0.00     0.12   0.36    0.00
## sex      0.45 0.00 0.09     0.53   0.01    0.04
## bmi      0.00 0.09 0.00     0.64   0.89    0.00
## children 0.12 0.53 0.64     0.00   0.78    0.01
## smoker   0.36 0.01 0.89     0.78   0.00    0.00
## charges  0.00 0.04 0.00     0.01   0.00    0.00
##
##  To see confidence intervals of the correlations, print with the short=FALSE option
```

### Regression Model Output

```r
library(olsrr)
MultiRegMedCosts.model <- lm(charges ~ age + sex + bmi
                             + children + smoker, data = MedCosts)
ols_regress(MultiRegMedCosts.model)
```

```
##                           Model Summary
## --------------------------------------------------------------------
## R                        0.866       RMSE                  6069.725
## R-Squared                0.750       Coef. Var               45.739
## Adj. R-Squared           0.749       MSE               36841564.604
## Pred R-Squared           0.747       MAE                   4178.656
## --------------------------------------------------------------------
##  RMSE: Root Mean Square Error
##  MSE: Mean Square Error
##  MAE: Mean Absolute Error
##
##                                        ANOVA
## -----------------------------------------------------------------------------------
##                          Sum of
##                         Squares          DF        Mean Square       F         Sig.
## -----------------------------------------------------------------------------------
## Regression    147001257515.223            5    29400251503.045    798.019    0.0000
## Residual       49072964053.144         1332       36841564.604
## Total         196074221568.367         1337
## -----------------------------------------------------------------------------------
##
##                                         Parameter Estimates
## -----------------------------------------------------------------------------------------------------
##       model           Beta    Std. Error    Std. Beta         t        Sig          lower          upper
## -----------------------------------------------------------------------------------------------------
## (Intercept)     -12052.462       951.260                 -12.670    0.000     -13918.594     -10186.330
##         age        257.735        11.904        0.299     21.651    0.000        234.383        281.087
##         sex       -128.640       333.361       -0.005     -0.386    0.700       -782.609        525.329
##         bmi        322.364        27.419        0.162     11.757    0.000        268.576        376.153
##    children        474.411       137.856        0.047      3.441    0.001        203.973        744.849
##      smoker      23823.393       412.523        0.794     57.750    0.000      23014.126      24632.659
## -----------------------------------------------------------------------------------------------------
```

## Collinearity Diagnostics

```
ols_vif_tol(MultiRegMedCosts.model)
```

```
##   Variables Tolerance      VIF
## 1       age 0.9850969 1.015129
## 2       sex 0.9911997 1.008878
## 3       bmi 0.9856317 1.014578
## 4  children 0.9977630 1.002242
## 5    smoker 0.9935846 1.006457
```